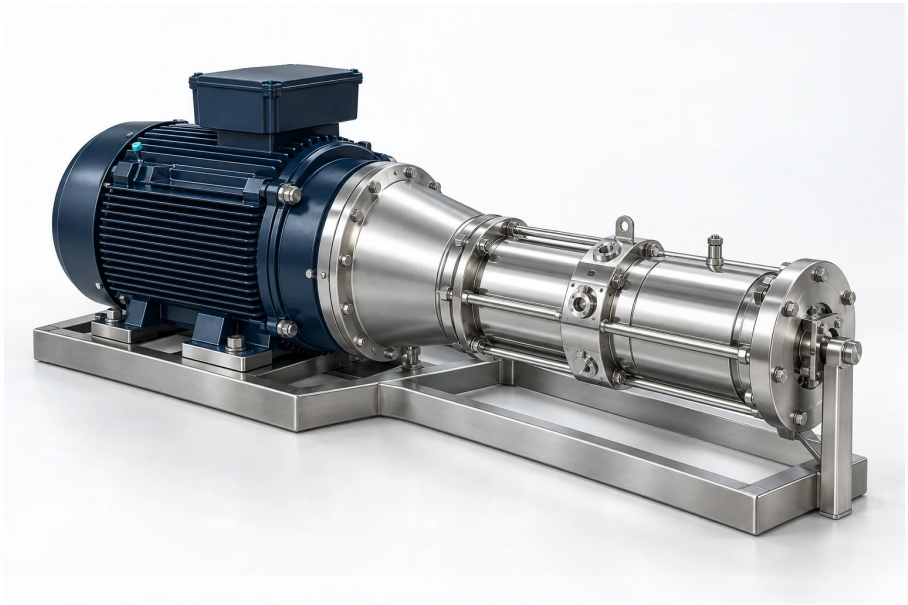
POWER CENTRE REVERSE OSMOSIS / PRODUCT INFORMATION

PC500: feed pumping and brine energy recovery

Compare the PC500 configuration at the required RO feed flow, pressure and recovery. Check feed and brine paths, motor input and any booster duty together; reference curves do not guarantee membrane-system production.

Bring high-pressure pumping and energy recovery together for larger seawater reverse osmosis duties. The PC500 combines duplex construction with production up to 500 m³/day (132,086 US gal/day).



Recreated illustration of the PC500 Power Centre: electric motor, high-pressure pump, integrated energy recovery motor and mounting frame

Key operating parameters

Parameter	Published value
Configuration	Axial piston pump + integrated energy recovery motor
Housing, shaft and pistons	Duplex stainless steel
Operating medium	Sea / salt water
Maximum production	Up to 500 m³/day (132,086 US gal/day)
Operating pressure	30-85 bar (435-1,233 psi) listed; confirm upper limit

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Parameter	Published value
Configuration	Axial piston pump + integrated energy recovery motor
Housing, shaft and pistons	Duplex stainless steel
Operating medium	Sea / salt water
Maximum production	Up to 500 m³/day (132,086 US gal/day)
Operating pressure	30-85 bar (435-1,233 psi) listed; confirm upper limit
Minimum inlet pressure	3.5 bar (50.8 psi) at pressure switch port
Maximum operating temperature	50 °C (122 °F)
Shaft speed	500-1,250 rpm
Mounting	Horizontal; support rear of unit
Drive connection	Splined shaft and flexible gear coupling
Power rating	Duty dependent - see performance curves
Filtration requirement	Confirm micron rating and water-quality limits

PC500 connections · Technical reference p. 9

Port	Function	Connection
A	Low-pressure out - energy recovery motor	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
B	Low-pressure in - pump	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
C	High-pressure in - energy recovery motor	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
D	High-pressure out - pump	3" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
E	Case drain - pump	1½" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
F	Case drain - energy recovery motor	1½" SAE 4-bolt flange, 3,000 psi (207 bar (3,002 psi)) class
G	Inlet pressure switch	G¼" BSPP female

PC500 recovery & ordering references

Recovery	Order code
30%	617-001
35%	618-001
40%	619-001
45%	620-001
50%	612-001

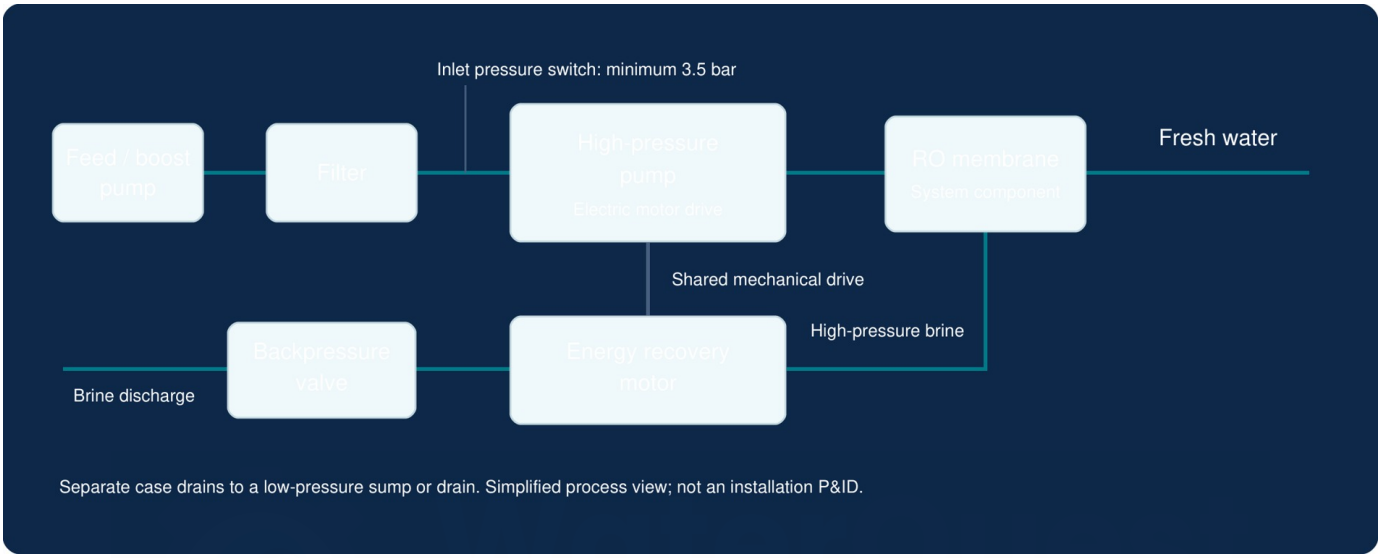
Selection and integration

Engineering checks Confirm before selection Pressure rating. The specification table lists 85 bar (1,233 psi) maximum; the circuit description states 80 bar (1,160 psi). Confirm the approved maximum for the selected unit. Recovery options. The ordering table lists 30-50% recovery. The narrative also mentions 25%; availability and an order code at 25% need confirmation. Production and power. Confirm duty at your feed salinity, temperature, membrane selection, recovery and operating hours.

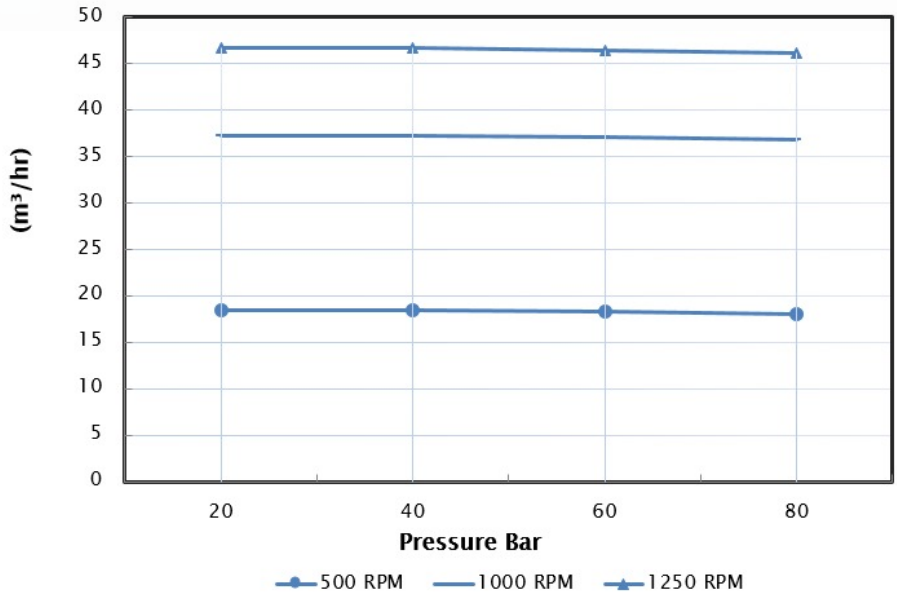
Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

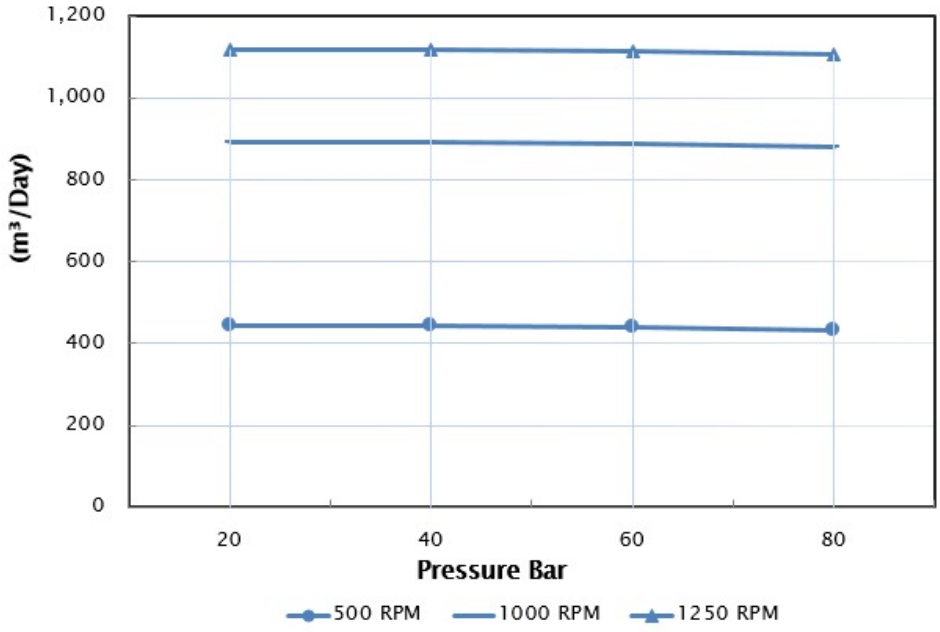
Process diagram: filtered feed enters the high-pressure pump, the RO membrane produces fresh water, and reject brine returns through the energy recovery motor to assist the shared drive.



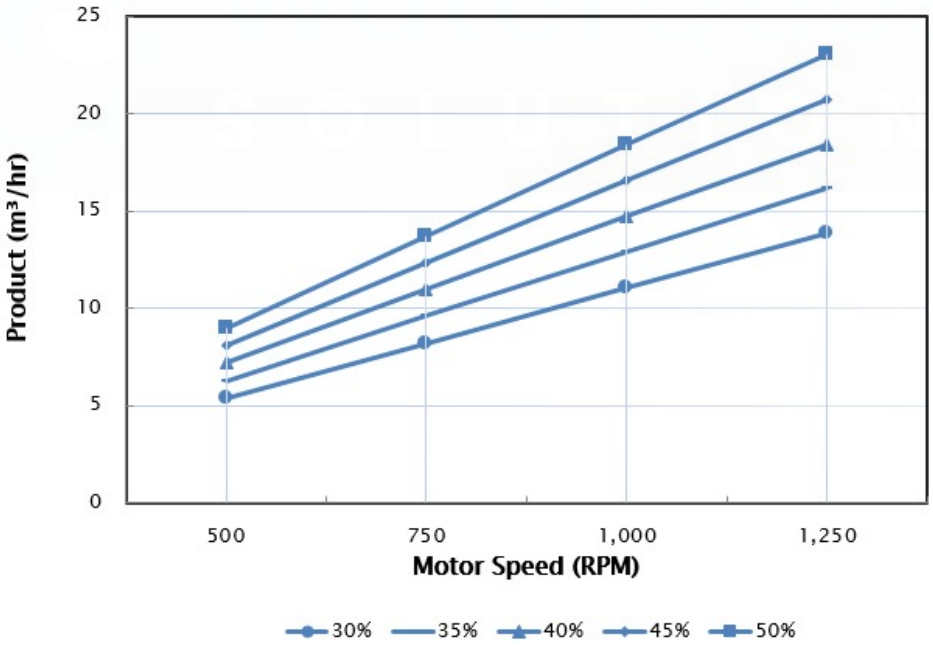
PC500 pump output flow chart, m³/h. Flow varies with shaft speed and operating conditions.



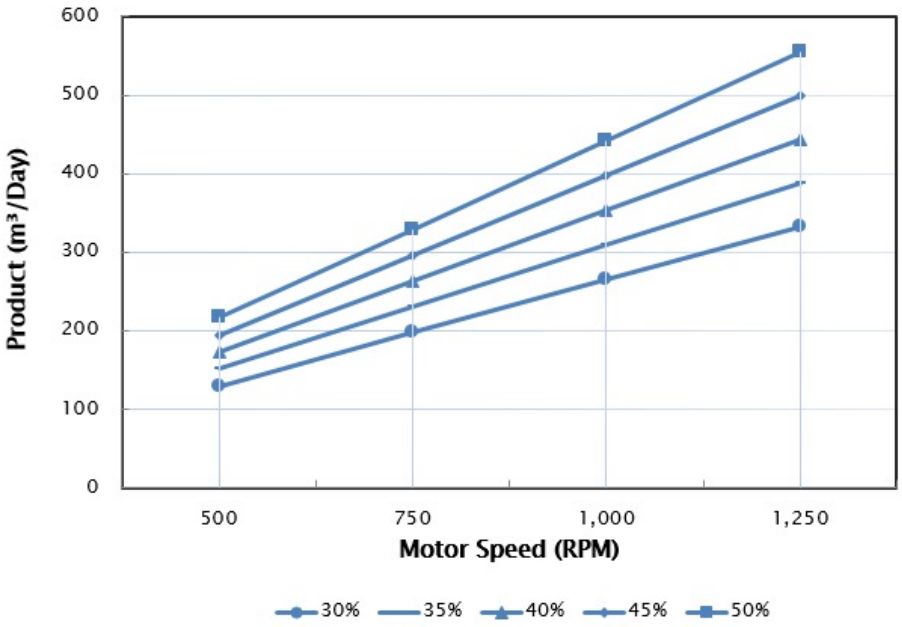
PC500 pump output flow chart, m³/day. Flow varies with shaft speed and operating conditions.



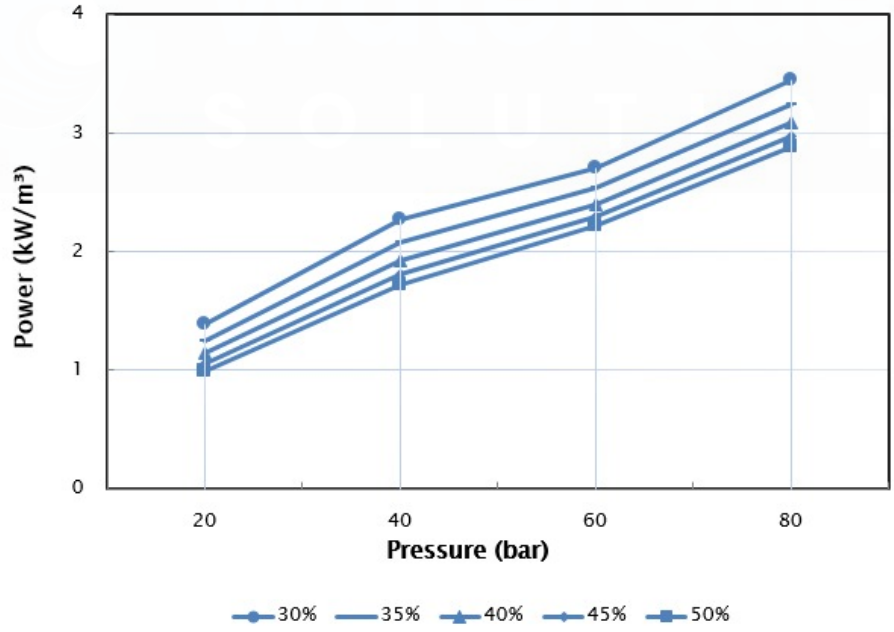
PC500 freshwater production chart, m³/h. At 80 bar; recovery selected when ordering.



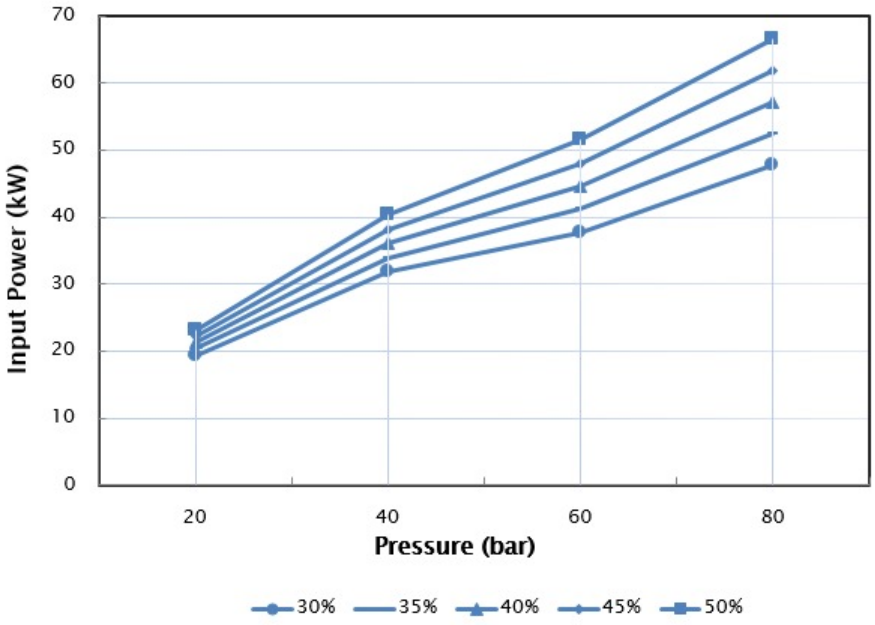
PC500 freshwater production chart, m³/day. At 80 bar; recovery selected when ordering.



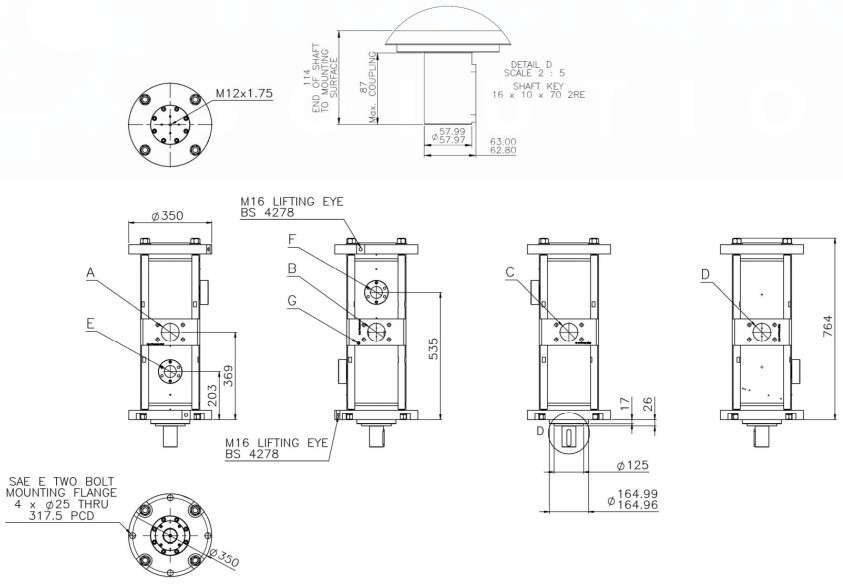
PC500 specific energy chart, Chart units: kW/m³. Performance curves at 1,250 rpm.



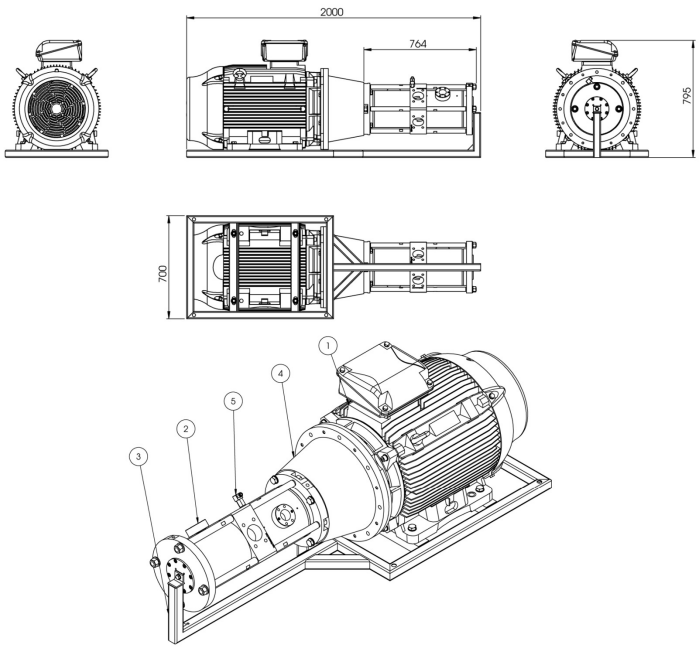
PC500 input power chart, kW. Performance curves at 1,250 rpm.



PC500 dimensioned unit drawing with port positions; open full-size for dimensions



PC500 motor, coupling, Power Centre and mounting frame assembly drawing



PC500 circuit schematic

